4. **Reaming of Flue Sheets:** When a flue hole in a fire box sheet becomes ⅛ inch out of round, it must be reamed.

6. **Preparation of Smoke Box Tube Sheets:** Unless otherwise shown on standard tracings, the diameter of the holes in the smoke box flue sheet must be 1/16 inch larger than the flues whose diameter is 2½ inches or less, and 3/32 inch larger than the flues whose diameter is 2¾ inches or more.

All holes must be cleaned and sharp edges removed. In superheater locomotives, the outside edges of the two rows of flue holes just below the lower row of large superheater flues and the outside edges of all flue holes above these two rows, must be rounded to 1/16" radius as shown in Sketch "A", since all these flues must be beaded at the smoke box end.

7. **Preparation of Fire Box End of Flues:** Fire box end of flues must be swaged down with a slight taper toward the end of the flue so that the diameter at the end will be .150 inch less than the diameter of the flue hole in order to permit the end of the flue to fit the ferrule neatly after the ferrule has been tightened in the flue hole.

The swaged ends of all flues, except the large superheater flues, must be at least 1⅛ inches long as shown in Figure 3. The swaged ends of large superheater flues must be at least 2 inches but not more than 16 inches in length as shown in Figure 10. The fin which is formed on safe ends when they are cut, must be rolled flat when the safe ends are used at the fire box end of the flues.

8. **Preparation of Smoke Box End of Flues:** Flues should be cut to length and the smoke box end opened hot to such a size that when placed in boilers they will drive snug in front flue sheet. This should not be done to flues which are to be transferred in the boiler.

9. **Tools for Working Flues:** All tools for working flues or arch tubes in locomotive boilers must be in accordance with standard tracings and the tool numbers stamped on each tool with ¼ inch figures. In addition, tools Nos. 4 and 8 must be stamped to show the thickness of the flue sheet with which they are to be used. All tools No. 5 must be maintained within limits of standard gauges, and tools Nos. 4 and 8 must be taken out of service when worn or distorted to such an extent that the tool will not force the flue sqaurely agains the sheet.

APPLICATION OF FLUES IN SATURATED STEAM LOCOMOTIVES AND THE SMALL BOILER FLUES IN SUPERHEATER LOCOMOTIVES

10. **Operation No. 1:** Ferrules must be placed in flue holes in fire box flue sheet as shown in Fig. 1, and rolled tight with tool No. 1, or expanded with tool No. 2.

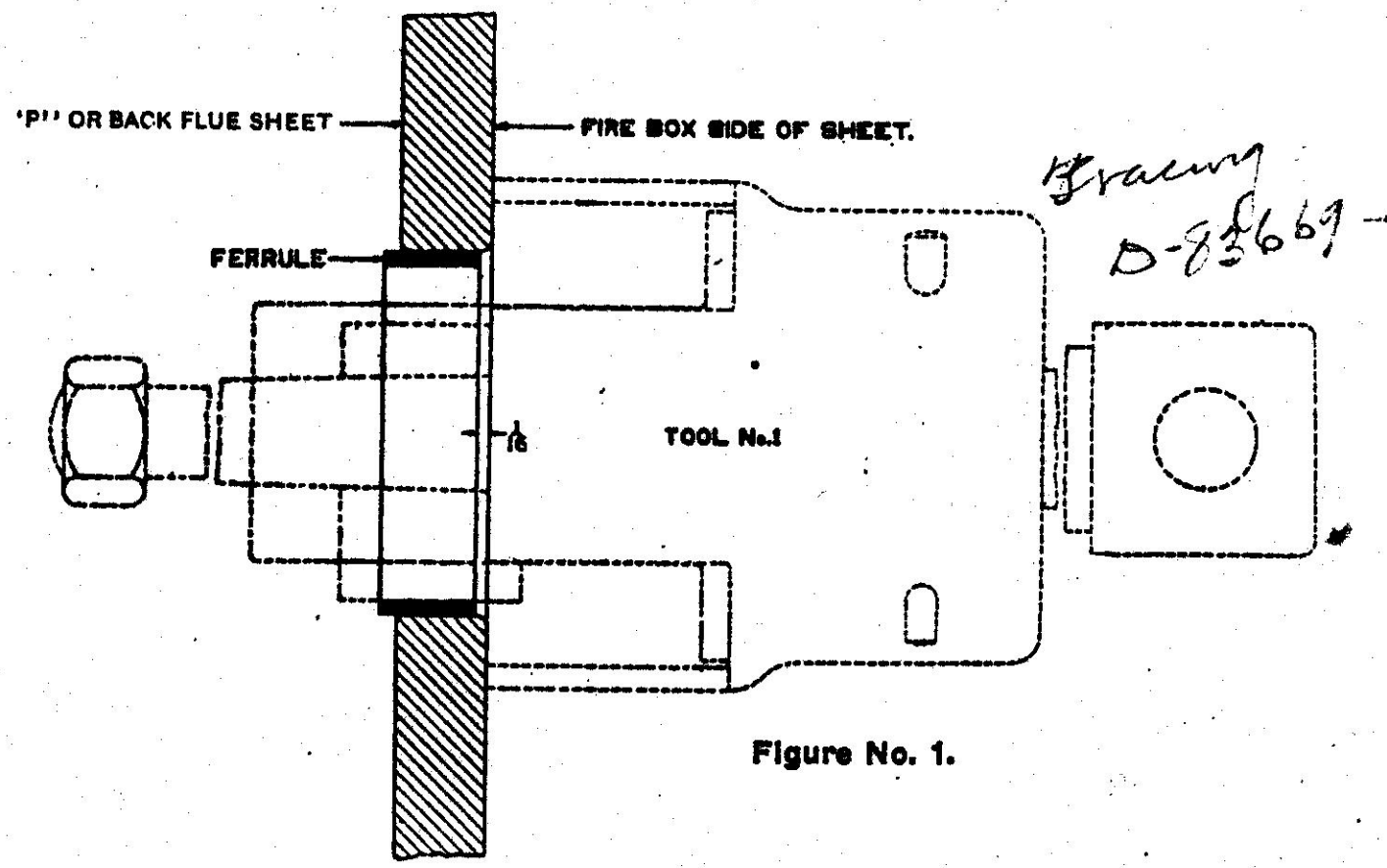


Figure No. 1.